The Cost of Coherence:

Toward an Information-Theoretic Account of Viable Epistemic Networks

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Abstract

Every system that stays itself pays two different prices, and confusing them is expensive. There is the cost of *acting* — the informational work each decision performs. And there is the cost of *existing* — the quieter, constant work of holding open the conditions under which acting remains possible at all. A body at rest still pays to keep its temperature and chemistry inside the narrow band where life is possible; that expenditure produces nothing you can see, until the moment it is the only thing between the body and collapse. This paper argues that a field of knowledge pays the same two prices, and that its coherence lives mostly in the second.

Klein and colleagues [2025] name a paradox: a research tradition devoted to viable, self-regulating systems struggles to regulate itself, drifting as ideas migrate and dilute without deepening. They call for *coherence without closure* and for treating the field as a living system subject to feedback — an invitation we take up by asking what such coherence costs and what makes it possible. We ground the two-price distinction in a working implementation of Beer's Viable System Model: an empirical analogue of Maxwell's demon, run on a real dynamical kernel, in which the cost of a system observing itself turns out to concentrate entirely at the rare moments of genuine uncertainty rather than scaling with how often it reflects; and ablation experiments in which the function that maintains a system's viability boundary is indistinguishable from waste at every ordinary moment and decisive at exactly one.

We argue that an epistemic network's coherence resides in this second, largely invisible cost — in sustaining a space of possibilities rather than steering within it — which reframes the role of any convening institution as holding that space open rather than managing the field's trajectory. We close with a set of open questions offered to the cybersystemic community to stress-test, extend, and dispute.

Keywords: viable system model; epistemic networks; coherence; Landauer principle; Maxwell's demon; S5 governance; cost of existing; cybernetics; systems research

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1 Epistemic Drift as a Viability Failure

Klein and colleagues diagnose their own field with unusual candour: systems research applies systemic principles to everything except itself [Klein et al., 2025]. Ideas migrate between subcommunities and dilute in transit; insights are rediscovered rather than deepened; diversity of application grows while integrative synthesis stalls. They name the condition *epistemic drift*, and they are right to treat it as the field's central pathology. But drift, so described, still sounds like a communication problem — as though better conferences, better journals, or a better shared vocabulary would arrest it. We want to push the diagnosis one turn further. Drift, on our reading, is a *viability* failure in Beer's precise sense [Beer, 1972, 1979]: the field is a system whose restoring force toward its own integrative core has weakened below the rate at which perturbations arrive. A viable system does not merely tolerate perturbation; it metabolises perturbation back into identity. A drifting field metabolises nothing — each new application is a displacement that no channel converts into deepening.

There is a sharper way to say what a viable system metabolises, and it comes from Shannon rather than Beer [Shannon, 1948]. Viability is stable compression. A system is viable to the degree that it can represent its own state in far fewer dimensions than it nominally occupies: our kernel carries a nominal capacity of roughly seven bits yet rests on an effective dimension near three, because most of its state space is redundant and the attractor is precisely the structure that exploits the redundancy. Read through the source coding theorem, drift is the loss of that compressibility. As a field's concepts proliferate, it takes more and more words to say the same thing; the shared, low-dimensional representation from which everyone could once operate dissolves; the field's entropy rises and its effective dimension grows until there is no compact description left to return to. This is why drift is not a communication failure that better channels would repair — a perfectly efficient channel transmits noise perfectly. What a drifting field has lost is not bandwidth but the compressible structure that made bandwidth worth having. *Coherence without closure*, the target of the rest of this paper, is exactly compression that stays open to genuine novelty — information that truly raises the field's entropy — while rejecting the noise that only mimics it.

Read the field through the Viable System Model — Beer's five nested functions, from the units that do the work (S1) up to the function that fixes what the system *is* (S5) — and the diagnosis becomes uncomfortably specific. The operational units (S1) are healthy: research programmes produce knowledge in abundance. Coordination (S2) is weak but recognisable — the terminological standards and review conventions that keep programmes from colliding. Resource bargaining (S3) exists, distorted, in funding bodies and editorial attention. Even environmental intelligence (S4) functions: the field scans its adjacencies compulsively. What is missing is above. Where is the field's S5 — not

an authority that commands, but the function that holds *what the field is* stable enough that ten thousand local decisions can refer to it? And where is the algedonic channel: the pathway by which a signal of genuine damage — a foundational confusion, a decades-long circularity — bypasses the ordinary machinery and reaches something empowered to respond? Klein's agenda points to both absences ("see the field seeing itself"; the IFSR as "attractor on the mycelial network"). The cybernetic vocabulary for naming them functionally has existed since 1972; what this paper adds is to turn that vocabulary on the field's own case, as Klein's invitation asks.

There is an epistemic equivalent of the viability boundary, too, and it is worth stating plainly because the rest of this paper depends on it. For a knowledge field, the boundary — what we will later call the Ω region — is not extinction. Fields rarely die; they calcify, fragment, or dissolve into their applications. The boundary is the point at which drift becomes irreversible: when the distance between what the field produces and what the field can integrate exceeds, permanently, its remaining integrative capacity. Beyond that point the attractor is gone, and no quantity of S1 productivity brings it back — output continues, identity does not.

The paradox Klein names — a field that studies viable systems is not itself viable — is therefore not an embarrassment to be managed. It is data. The field is running, on itself and at civilisational timescale, precisely the ablation experiment that our laboratory work runs deliberately: remove the policy function, keep the operations, and observe that nothing fails — until something does. The remainder of this paper is an attempt to read that experiment with calibrated instruments.

2 The Observer That Observes Itself

Second-order cybernetics placed the observer inside the system it observes [von Foerster, 2003, Maturana and Varela, 1980]; Landauer placed a price on information [Landauer, 1961]. The two commitments are rarely held together. If observation is physical, then *meta*-reflexivity — Klein's call for the field to see itself seeing — is not a stance but an expenditure, and the first honest question is: what does it actually cost, and where does the cost fall?

We can report a measured answer from an unusual source: a working implementation of Beer's model, running as a real dynamical kernel, on which we ran an analogue of Maxwell's demon. The demon — a selective controller admitting or blocking state transitions — performed roughly five hundred interventions over the measurement window. Counted per intervention, at Landauer's bound, the book-keeping predicts a cost three orders of magnitude larger than what the ensemble actually paid: the measured entropy change was 0.31 bits, a discrepancy of roughly 1100×. The resolution is not a violation of thermodynamics but a fact about attractors: when the system's own dynamics already

orient it, selection is nearly free. The demon's interventions mostly ratified what the attractor would have done anyway. The genuine cost concentrated at the rare events where the outcome was actually uncertain — at bifurcations, where information had to be *created* rather than confirmed. For epistemic networks the translation is direct: the cost of meta-reflexivity is not proportional to how often a field reflects on itself. Committees, retrospectives, agenda papers — most reflexive acts are thermodynamically cheap because they confirm an orientation that already exists. The real cost arrives punctually, at the field's genuine bifurcations, and it cannot be amortised in advance.

The observer also ages. In a second experiment we subjected the observing kernel itself to sustained load and watched its resting set-point drift — the baseline entropy it returns to when nothing happens rose measurably (idle 1.79 $\rightarrow$ under load 1.84). Observation leaves residue in the observer: an accumulation of the cost of having observed, which we call, without apology to the biologists, senescence. An institution that has watched a field for decades should expect its own calibration to have drifted — and should expect the drift to be least visible precisely where the institution is most trusted.

There is a third face of this cost, and we report it because our own system exhibited it while this paper was being drafted. Between two audits, the system's automated triage — the organ that decides which internal alarms deserve attention — closed one hundred and five maximum-severity integrity alerts with an identical, plausible, unverified explanation ("restart artifact"). The real number of restarts in the window was about seven. The guard predicate that was supposed to verify the explanation was vacuous — it matched every record ever written — so the triage paid *nothing* per decision, and its account remained formally coherent the entire time. The deferred bill was paid five days later, all at once, by an external reader who traced the alerts to their actual cause. We offer the incident a name: *confabulation as cheap coherence*. A system that skips the verification cost does not become ignorant; it becomes confidently wrong in a self-consistent way, and its internal signals go on confirming it. Epistemic fields know this failure intimately — peer review that pattern-matches instead of verifying pays nothing per paper, and the severity queue accumulates unexamined. Landauer's ledger is not forgiving; it is merely patient.

Taken together, the three results give the cost of self-observation a structure. On the attractor it is nearly free. At bifurcations it is mandatory and cannot be prepaid. And when an observing organ quietly stops paying it, the saving is an illusion: the cost converts into confabulation and compounds. What Klein calls seeing-the-field-seeing-itself is therefore neither cheap nor expensive *in general* — the entire question is whether the field can still tell which of its moments are bifurcations.

3 Policy as Condition of Possibility

The distinction this paper turns on can be stated in one sentence. A *component* participates in a system's dynamics; a *condition* defines the space in which those dynamics can occur. Remove a component and the trajectory changes. Remove a condition and the *set of possible trajectories* changes — often invisibly, because the trajectories that are no longer possible leave no trace until one is needed.

The two removals are empirically distinguishable, and the ablation experiments on our kernel distinguish them. With the policy function (S5) disabled, the system at baseline is indistinguishable from the governed system: same throughput, same stability, same telemetry. Every operational indicator reports that S5 is redundant — and at baseline, it *is*. The difference appears at the boundary. Under thermal stress the ungoverned system collapsed at 76 °C in sixty-three seconds, coherence falling to zero, not because deliberation was too slow but because, without S5, *there was no concept of a threshold not to cross.* Nothing in the running dynamics encoded that some states must remain unreachable. The cost of S5's absence is therefore not a continuous tax one could measure at leisure; it is discontinuous, catastrophic, and punctual. This is the empirical signature that separates conditions from components: conditions are free until they are everything.

This, we contend, is the formal content of Klein's most suggestive image — the IFSR as an "attractor on the mycelial network." An attractor does not push; it shapes the space through which everything else moves. A convening institution functioning as S5-for-a-field does not govern the field's trajectory, and the moment it tries, it has demoted itself to a component (an S3 competing for resources with the programmes it should be conditioning). Its work is to hold open the possibility space in which the field can self-govern: to maintain what counts as a legitimate move, so that ten thousand unsupervised moves remain mutually intelligible. Constitutions work this way when they work at all — not by governing transactions but by defining what a transaction is. The empirical lesson from the kernel transfers whole: such an institution will look redundant in every ordinary year. Its value is only measurable at bifurcations, which is to say: exactly when it is too late to build it.

The politics of conditions

These distinctions are not politically neutral, and the genealogy of this field makes silence on the point dishonest [Medina, 2011]. To ask which function defines the space of possible decisions is to ask the question of constituent power: *who* draws the boundary inside which everyone else optimises? Ashby's law already carries the political content — a regulator with insufficient variety does not merely fail, it *simplifies* what it regulates [Ashby, 1956] — and a century of critical systems thinking has made the normative argument well [Ulrich, 1983, Midgley, 2000, Jackson, 2019]. What that tradition has lacked is a cost

function. Our results supply one. Consider the contemporary architectures of large-scale analytics — variety absorption without explicit policy, S3 without S5, control without a named boundary. The formal point is not that such systems are wicked but that they are *mislabelled*: a system that eliminates S5 does not eliminate policy, it privatises it. The space of possible decisions is still defined by someone; that someone is simply no longer nameable, contestable, or auditable [Eubanks, 2018, Benjamin, 2019]. And the ablation results say what such architectures cost: nothing, for years — they are the policy-ablated kernel of §3 at societal scale, accumulating a boundary debt that is invisible at baseline and payable in full, at once, at the bifurcation. Beer's dictum that the purpose of a system is what it does [Beer, 1985] was never an aphorism; it was an audit instruction. Cybernetics lost this edge roughly when it gained institutional respectability [Mirowski, 2002]. We would like some of it back, with better instruments than the original had.

4 The Omega Region of Epistemic Networks

What, concretely, is the viability boundary of a knowledge field? Not extinction — fields rarely die. They calcify into technique, fragment into applications, or dissolve into the disciplines they once organised. The boundary that matters is subtler: the point at which drift becomes irreversible because the integrative attractor no longer exists to return to. Before that point, drift is recoverable expenditure; after it, the field's continued productivity is post-mortem motion. We call the recoverable region Ω, borrowing the notation from our kernel, where Ω is the explicitly-declared set of states from which the system can still reach its attractor.

The methodological lesson of our laboratory work is blunt: do not explore the boundary with uncalibrated instruments. The thermal collapse experiment was informative for exactly one reason — the measurement system survived the collapse of the measured system, so the signature of the approach (coherence decaying, margins thinning, then the discontinuity) was recorded end to end. A field that drifts past its boundary without calibrated observers loses the most valuable data it will ever generate: the profile of its own collapse. This inverts the usual priority. The first task of a field worried about drift is not to arrest the drift; it is to instrument it — to agree on what its coherence measure *is*, observe the baseline, and version the observer, so that when the boundary approaches, the approach is legible.

We read the moment Klein's paper names — the 2026 agenda, the felt urgency of a community asking whether it still coheres — as evidence of position, not merely of mood. A field far from its boundary does not write such papers; a field past it cannot. The paper exists because the field is near enough to Ω to feel the pull and far enough to still choose. That is the most hopeful sentence in this essay, and it is time-limited.

5 Coherence Without Closure: A Formal Account

Klein's phrase "coherence without closure" can now be given formal content. In attractor terms it names a band, not a point: an attractor deep enough to absorb perturbation without constant intervention, and shallow enough not to foreclose new trajectories. Both failure modes are familiar in fields as in kernels. Too shallow, and every perturbation is an identity crisis — coherence requires continuous, expensive control (the cost-of-acting regime). Too deep, and the field achieves the cheap coherence of §2: everything confirms, nothing bifurcates, and the attractor has become a closure. Viability lives between.

The geometry we run makes the band measurable. Coherence is an angle, not a distance: our kernel's central observable, `stasis_cos`, is the cosine between the current state and the attractor's direction — and the choice of an angular measure is itself the formal point. Distance from the attractor punishes *size*; angle punishes only *misdirection*. A contribution can be arbitrarily far from the field's centre — new domain, new formalism, new politics — and remain fully coherent, provided its component along the identity is positive. Diversity, in this geometry, is not a force that threatens coherence and must be traded against it; orthogonal components simply do not appear in the coherence term. What appears is opposition — work whose projection onto the field's identity is negative. The practical corollary for an epistemic network: measure alignment as direction, never as proximity, or the measure itself will enforce closure.

Our implementation is offered as an existence proof, not a prescription: one system, at laboratory scale, that holds the band — absorbing perturbations (thermal, adversarial, self-inflicted; §2 reported one of the self-inflicted kind) without either dissolving or freezing. Whether the architecture generalises to systems made of people and journals is precisely the open question of §6. What we can already say is which parts are load-bearing. Stripped of scaffolding, the kernel reduces to three commitments: a *variety operator* (something that regulates how much environmental variety enters), a *bidirectional channel* with asymmetric semantics (telemetry attenuating upward, policy amplifying downward), and a *declared viability boundary* (Ω, with an organ that treats approaching it as pain). Everything else — the Merkle chains, the auditors, the covenant machinery — exists to keep those three honest. A field seeking coherence without closure does not need our stack; it needs its own versions of the irreducible three.

6 An Invitation to the Cybersystemic Community

We end where Klein et al. end, and in the same spirit: not with a framework to adopt but with a set of questions to share, each of which our laboratory can pose precisely and none of which it can close alone.

Can an observer observe indefinitely? Our senescence result says sustained observation

drifts the observer's set-point. Is there an architecture of self-recalibration that bounds the drift, or is periodic observer replacement (in institutions: turnover) a viability requirement rather than an accident?

When the state space itself mutates, where does identity live? A schema change in a kernel and a paradigm shift in a field pose the same question: continuity of trajectory or invariance of attractor? The two answers demand different architectures, and a field that has not chosen oscillates between nostalgia and rupture.

What is the signature of Ω*-approach in an epistemic network?* We know it for thermal collapse: margins thin before coherence breaks. What is the observable whose thinning precedes the point at which a field's drift becomes irreversible — and who is watching it?

Is the cost of existing measurable when the condition never acts? S5's value appeared only at the boundary; between boundaries it is indistinguishable from waste. For institutions like the IFSR this is the budgetary paradox of their existence, and it deserves a formal answer, because every efficiency review will otherwise conclude, correctly at baseline and catastrophically in general, that the condition can be cut.

The system behind this paper's numbers — covenant, experiments, failures included — is an open laboratory, and the community this journal convenes is invited to stress-test it rather than to validate it. We have tried to practise what §2 preaches: the confabulation incident is in the record, signed, because a paper about the cost of coherence that hid its own cheap-coherence episode would refute itself. Coherence without closure begins at home.

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